

Quang Huy Pham

AI Engineer

huyqhkhd@gmail.com | [0332008242](tel:0332008242) | [LinkedIn](#) | [GitHub](#)

Experience

AI Engineer – VinSmartFuture

Nov 2025 – Present

- Developed FlowBuilder, an internal chatbot-building platform at VinBigData based on Langflow v1; built a custom MCP integration module using Server-Sent Events (SSE) to enable chatbot workflows to connect with external tools and services.
- Developed RagFlow, a retrieval platform that enables V-App to query Vingroup internal documents and accurately answer P&L-related questions; evaluated multiple reranking base models, built fine-tuning datasets from public and private sources, optimized the reranking pipeline, and achieved MRR@1 of 0.95 with 400 ms latency.
- Developed Agent-ingestion, a document parsing module that converts Vingroup internal PDF and DOCX files into clean Markdown for LLM ingestion; designed agent-based extraction pipelines, fine-tuned OCR and layout models for text and scanned documents, engineered prompts for structure recovery, and achieved WER of 0.01 for text PDFs and 0.03 for scanned PDFs.
- Developed V-App Guardrails, a production safety layer serving 180K requests/month; designed guardrail prompts, fine-tuned and benchmarked over 30 LLMs, deployed models on AWS across 20 GPUs, implemented daily tracking and monitoring across the model lifecycle, and achieved P90 latency of 500 ms at 200 concurrent users in production.
- Developed Vinhomes Lead Scoring, a machine learning system for prioritizing high-potential leads from a large pool of customers and project interests; consolidated customer data into a data lake, processed data into bronze, silver, and gold layers, performed EDA and feature engineering, trained and served models on Databricks, reducing sales leads by 30% while official deal conversions decreased by only 20%.

AI Engineer Trainee – VinBigData

Jul 2025 – Nov 2025

- Member of Cohort 5, Vingroup AI Engineer Training Program
- Gained a solid foundation in Machine Learning, Computer Vision, Natural Language Processing, Applied Mathematics, and Generative AI
- Enhanced supporting skills in Python programming, MLOps, data engineering, soft skills, and advanced technical seminars

AI Engineer – KienlongBank

Mar 2025 – Aug 2025

- Developed an internal HR chatbot serving over 3,000 employees.
- Researched and implemented chatbot architectures using Retrieval-Augmented Generation (RAG) and Knowledge-Augmented Generation (KAG).
- Utilized AI agents orchestrated through n8n to automate HR-related interactions.
- Analyzed large-scale banking transaction data to identify credit risk factors.
- Built AI agents to assess and monitor financial risk using rule-based systems and statistical modeling.

AI Engineer – DopikAI

Sep 2024 – Nov 2024

- Researched and fine-tuned large language models for internal deployment on GPU servers.
- Designed optimized prompts to enhance response quality and consistency.
- Developed scalable APIs using FastAPI to integrate AI models into core products.

Research and Publications

Multimodal Research on Scene Text VQA

- Conducted research on the Visual Question Answering (VQA) task, focusing on the integration of image processing and natural language processing in the context of scene text.
- Created two new datasets for the Scene Text VQA task: **ViTextVQA**, manually annotated via crowdsourcing; **ViOCRvQA**, the largest VQA dataset in Vietnamese to date.

- Proposed a novel model named **VisionReader**, which integrates three image-processing modules and one language-processing module. The model outperforms state-of-the-art methods when applied to Vietnamese.
- Published two Q1 journal articles and one medical journal article in this field, and received two academic awards related to VQA. References available at: [Link](#).

Natural Language Processing (NLP)

- Focused on research topics such as fake news detection, legal document retrieval, and question answering in the legal domain.
- Created several domain-specific datasets, including a Vietnamese legal QA dataset and an educational dataset for distinguishing between AI- and student-written texts.
- Developed task-specific language models optimized for Vietnamese to achieve state-of-the-art performance across tasks.
- Published three papers in this field, including one in a top-tier (Rank A) venue, and received four academic awards related to NLP. References available at: [Link](#).

Large Language Models (LLMs)

- Conducted research on adapting and fine-tuning large language models (LLMs) for Vietnamese, focusing on domain-specific tasks such as legal question answering, hallucination detection, and dialogue systems.
- Built several domain-adapted datasets (legal, education, social media) for instruction tuning, evaluation, and hallucination analysis.
- Applied advanced techniques such as low-rank adaptation (LoRA), quantization, and retrieval-augmented generation (RAG) to optimize LLM performance under limited computing resources.

Projects

Legal Chatbot

Designed and developed an AI-powered chatbot to answer legal questions based on Vietnamese law. Built a high-quality vector database from legal documents and integrated Retrieval-Augmented Generation (RAG) with web search for accurate, context-aware, and up-to-date responses.

- **Frontend:** Developed a real-time chat interface using React TypeScript and WebSocket for smooth interaction.
- **Backend:** Built RESTful APIs with FastAPI, managed structured data with PostgreSQL, and used Qdrant as the vector store.
- **AI:** Collected and fine-tuned embeddings and LLMs on Vietnamese legal texts to improve domain-specific accuracy. Applied 4-bit quantization with Unsloth to optimize memory usage under limited resources. Leveraged vLLM for high-throughput inference, enabling faster response time for real-time chatbot deployment.

Education

University of Information Technology (UIT) – AI Engineering

2021 – 2025

- GPA: 3.4/4.0 – Thesis: 9.7/10
- TOEIC: 625 (Listening & Reading), 220 (Speaking & Writing)

Skills and Tools

- **AI/ML & LLMs:** PyTorch, TensorFlow, scikit-learn, Hugging Face, Transformers, Sentence Transformers, LangChain, vLLM, LoRA, quantization, RAG/KAG.
- **Model Optimization & Serving:** ONNX, Triton Inference Server, vLLM, Unsloth, Docker, MLflow, Databricks, AWS.
- **Data Engineering & Backend:** Python, Pandas, Spark, Airflow, Kafka, FastAPI, Django, PostgreSQL, MongoDB, Redis.

Competitions and Achievements

- **VLSP 2023:** Third Prize in Legal Textual Entailment Recognition; Top 4 Private Test in Visual Reading Comprehension for Vietnamese.
- **UIT Data Science Challenge:** First Prize (Group B) in Vietnamese Fact Checking, 2023; Top 1 and Top 3 in Group B, 2025.
- **DSAA 2023:** Top 4 Private Test in Link Prediction for Wikipedia Articles.